

I used AI a lot during my final project, but I'm going to break it down into what I thought was most important by section, coding, paper, and website. For the coding, I primarily used it as an error explainer and when I was confused on how to interpret the outputs of my code (primarily regression). The largest among of pure AI code I used was the kernel regression. I first tried to do it myself, and I couldn't figure it out, so I asked chat to fix it. It still made a lot of errors and things that I didn't like that I fine tuned with different prompts. For the paper, I mainly used it for pulling stuff from earlier assignments, proofreading after giving the rubric, and turning it into overleaf. Again it made a mistake and rewrote my entire paper while formatting it on overleaf, which I had to redo. Finally on the website, I used agentic claude and asked it to go through my folder, and create a website using the general format of a scientific paper. It messed up the colors and I ran out of credits so I asked chat to fix it.

3 Transcripts

Input:

"I can't figure out this Kernel regression, what should I do" *attached the Kernel regression code I currently had

Output:

"Here is a complete version, with these specific changes"

Reaction:

"It was super helpful, I used most of the code, but there were some issues with the smoothing of the code that I had to go back through and have a conversation with chat to fix."

Input:

"I made a mistake and I need to rewrite this paper in a specific overleaf format, can you transfer my work to this format" *attached format and work*

Output:

Chat gave me an overleaf template with what I thought was my work, but when I read it, it had basically rewritten everything, so I had to go back and copy and paste my original wording

Reaction:

Slightly worried me, but it ended up being super helpful because it showed me how to insert equations, so I went back and copied the equations that chat had inserted because they look super professional.

Input:

"I got the rubric, can you tell me which sections need to work on" *with attached rubric

Output simplified:

"I'd focus on Data, Methods, and your Repo"

“Data needs more clarification of where it is from and the time period”

“Methods needs more concrete coding of how things worked”

“Repo needs to be better cleaned and organized”

Reaction:

I listened to most of the suggestions, I had made some preliminary edits on my own but going to fine tune it with this.

Tool	Purpose	Examples
ChatGPT	Python coding/debugging	Kernel regression, plotting, treatment-group coding
ChatGPT	Statistical interpretation	Regression coefficients, transition curves
ChatGPT	Writing feedback	Organization, wording, lightning-talk preparation
Claude Code	Website development	React/CSS layout, formatting, debugging

Reflection:

AI definitely saved me a lot of time, especially with editing, I do worry that I'm becoming reliant on this aspect of AI that I know the basics and AI becoming a crutch for the more complex parts, but at the same time, I am still learning the complex parts. I accepted most of the error fixing, but sometimes chat would misinterpret my prompt and create a different output than I originally intended, so I had to go back and correct it. It was also somewhat misleading and would create new variables in the code that I hadn't used before or it mistakenly thought was in the dataset, so I frequently had to go back through the code and fix it. It was also too much of a “yes man.” When I originally started the cubic transition model, I asked chat how it was looking in comparison to the original paper, and it was way too enthusiastic, and failed to notice that I completely missed the productive assets part. It wasn't until I was going through all of the outputs that I noticed the difference. It ultimately was a small fix, but one that I thought it should have caught earlier.